

Aurela Broqi

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science, GPA: 3.31

Orlando, FL

Expected Graduation, July 2026

EXPERIENCE

Air Force Research Lab | Virginia Tech

- Benchmarking and Evaluating Uncorrelated Track Processing Algorithms

Orlando, FL

Incoming September 2025

M.E.T.I.L | Software Engineer Intern

Institute of Simulation & Training

- Deployed and containerized OpenWebUI using **Docker** on a high-performance host machine, configuring network access via host IP and port to provide centralized usage for developers and interns. Researched and tested AI model performance through architecture comparisons and quantization techniques to improve response time and accuracy.
- Collaboratively integrated a **Unity**-based VR application into the VIROO® platform to enable scalable deployment within an enterprise-grade immersive environment.
- Designed spline-based navigation paths and configured VR controllers to enable dynamic interaction with scene game objects.
- Implemented version control best practices using **Git/GitHub**, improving team collaboration and code integration.

Orlando, FL

May 2025 – Present

M.E.T.I.L | Backend Research Intern

Institute of Simulation & Training

- Analyzed and documented qualitative training requirements into measurable complexity metrics, supporting backend design planning for a simulation and **reducing manual tuning efforts by an estimated 50%**.
- Wrote technical documentation mapping system outputs to complexity factors, enabling adaptive performance scaling and improving **process alignment with user requirements by 30%**.
- Mapped backend logic using a RAG-based framework to adapt system workflows based on performance and complexity metrics, supporting future implementation of personalized, data-driven processes.

Orlando, FL

December 2024 – January 2025

PROJECTS

Sales Objection Handling AI Tool | React, Node.js, Express, Gemini AI API

Solo Project

- Developed a web app that transcribes speech and generates real-time, context-aware responses to client sales objections.
- Built with a React frontend and Node.js backend, integrating Express and the Gemini API for conversation analysis and response delivery.
- Implemented objection-handling logic using techniques such as urgency, need, scarcity, and assumptive close to improve call flow and engagement.

Orlando, FL

June 2025

World.ihardcodebubblesort.com | MERN Stack (MongoDB, Express, React, Node.js)

Group Project – Backend & Database

- Developed backend and database architecture for an interactive globe web app where users drop pins to log visited locations.
- Utilized VMware to deploy the project environment; built RESTful APIs in **Node.js/Express** for image uploads, geolocation, login, and registration.
- Created MongoDB schemas for user authentication and pin data, including fields for coordinates, image URLs, descriptions, usernames, and timestamps.
- Documented API endpoints, database schema, and project dependencies to support team collaboration and deployment.
- Collaborated in a team of 6 using **Jira** for planning and **Git/GitHub** for version control and code reviews.

Orlando, FL

April 2025

SimpleIR to x86 Compiler | C++ & ANTLR

Solo Project

- Developed a compiler that translates SimpleIR into x86 assembly, implementing parsing, semantic analysis, and code generation in a Linux environment using CLI tools and Git for version control.
- Utilized ANTLR to define grammar and generate an abstract syntax tree for structured compilation.
- Implemented register allocation, branching logic, and pointer operations to enhance execution efficiency.

Orlando, FL

February 2025

TECHNICAL SKILLS

- Programming Languages:** Java, C, Python, HTML, CSS, C++, C#, PHP, JavaScript, Shell Scripting, Node.js.
- Software & Tools:** Anaconda, AWS, Unix/Linux, VS Code, Tomcat, GitHub, Jira, Docker, VMware, VS Code, Postman, MySQL, Bootstrap, Unity, Unreal, SteamVR, Vive, Apache, MongoDB, VIROO, OpenWebUI, MCP.